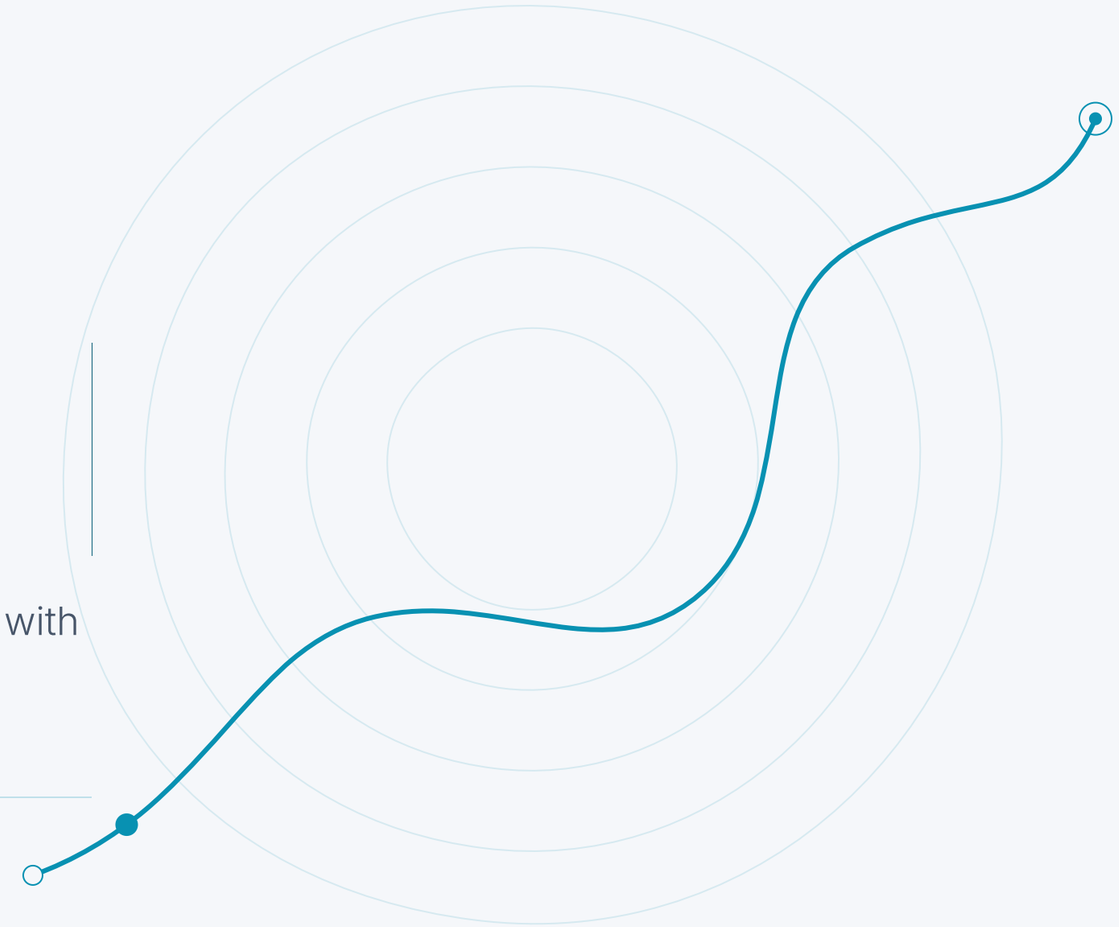
SE322 · SOFTWARE DESIGN & ARCHITECTURE · SECTION 1340

Intaliq

The architecture of an outdoor fitness app for Saudi Arabia, designed with the 4+1 View Model.



INSTRUCTOR

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PREPARED BY

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Rose Al Rakan
Mozoun Almuqrin
Shoug Alomran
Munira Alfaris

OUTLINE

What we'll cover

From the problem Intaliq solves to the architecture behind it, the design patterns that shape the code, and how we validated the design.

01 **Introduction**
Problem, purpose, scope and key definitions →

02 **4+1 Architecture**
Goals, constraints and the five architectural views →

03 **Quality Attributes**
Security, availability, performance and usability →

04 **Architectural Style**
Client–server, and why it fits Intaliq →

05 **Software Design**
MVC and layered views; Factory Method, Observer and Adapter →

06 **Validation**
Requirement coverage and design review →

DELIVERABLE 1 · 40%

Software Architecture Design

✓ SUBMITTED 30 SEP

4+1 VIEW MODEL

8	5	11	3
SAD sections	architectural views	UML diagrams	use-case scenarios

01

SAD 1.1 · OVERVIEW

Getting active outdoors is harder than it should be

Monitoring outdoor activity

Few tools accurately track how active people are outdoors.

Finding partners

Many struggle to find someone to join their preferred activity.

Joining group sessions

Large, professionally led group events are hard to find in Saudi Arabia.

VISION 2030 · QUALITY OF LIFE TARGETS

Population exercising weekly

13% → 40%

Average life expectancy

74 → 80 yrs

Physical activity is key to reducing chronic disease — Intaliq helps citizens adopt a more active, healthier lifestyle.

SAD 1.2 · PURPOSE

Make running, cycling and hiking accessible and enjoyable

Intaliq is a mobile fitness app. Users track daily activity, set goals and join group sessions held daily by a trained team of coaches, through a simple design that makes outdoor activity feel less intimidating.

01

Easy navigation

Guide users through outdoor activities in Saudi Arabia with an easy-to-use interface.

02

Accurate tracking

Provide reliable tools for monitoring activity and setting activity goals.

03

Social connection

Connect people across Saudi Arabia who share an interest in outdoor activity.

SAD 1.3 · SCOPE

Eight features, two roles

A smartphone app that raises the accessibility, motivation and social side of outdoor fitness through tracking, connections and organised sessions.

USER

Activity tracking

Log daily outdoor activities and track improvement.

USER

Goal setting

Set and adjust activity goals to monitor progress.

USER

AI analysis

Chat with an AI chatbot for an overview or goal tweaks.

USER

Finding partners

Chat and pair up with partners to stay motivated.

USER

Join group sessions

Running, hiking and cycling crews hosted by Intaliq coaches.

COACH

Create sessions

Create and schedule new group sessions.

COACH

Admit members

Accept applicants whose goals and level match the session.

COACH

Announce updates

Notify members ahead of time about session changes.

SAD 1.4 – 1.5 · DEFINITIONS & REFERENCES

The data dictionary behind every view

Session — CLASS

date	DD/MM/YYYY, realistic
time	24-hour clock
location	In KSA, permitted for groups
capacity	> 1, fits the venue
intensity	Beginner · Intermediate · Advanced

User — CLASS

firstName	< 20 characters
lastName	< 35 characters
phone	+966 5X-XXX-XXXX or country code
email	Recognised domain, handle < 64
password	≥ 8, upper case, number, symbol

Coach — CLASS

name	First and last < 20 characters
phone	Same rules as User
email	Same rules as User
password	Same rules as User

Goal — CLASS

type	Hiking · Biking · Running
targetTime	HH:MM
calories	> 10 kCal
deadline	DD/MM/YYYY, not in the past

Partner — CLASS

location	In KSA, permitted for groups
goals	One of the three activities
activities	Hiking · Biking · Running

Activity — CLASS

type	Hiking · Biking · Running
duration	5 minutes to 1 day

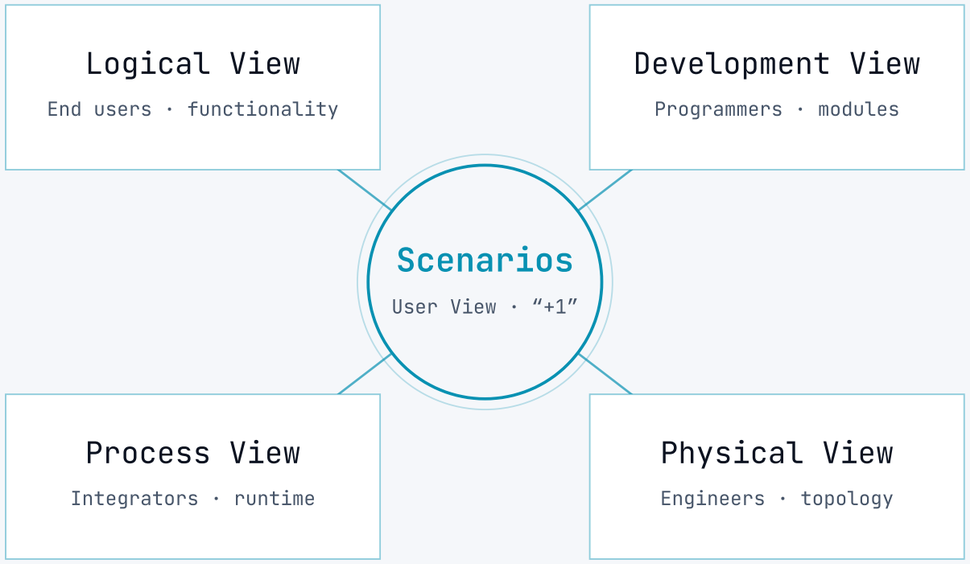
1.5 REFERENCES

- INTALIQ VISION & SCOPE DOCUMENT
- IEEE SRS TEMPLATE
- INTALIQ 2.0 SRS · SE311 V1.0

SAD 2 · ARCHITECTURAL REPRESENTATION

The 4+1 View Model

Philippe Kruchten (1995). Users, coaches and developers each see Intaliq differently, so the architecture is split into four views, tied together by scenarios — the “+1”.



+1 · USER VIEW

Use-case diagram and three scenarios

LOGICAL

Class diagram and five sequence diagrams

PROCESS

Two activity diagrams — user and coach

DEVELOPMENT

Component diagram with interfaces

PHYSICAL

Deployment diagram — device, server, database, AI

SAD 3 · ARCHITECTURAL GOALS & CONSTRAINTS

What the architecture must protect and respect

GOALS

Easy-to-use mobile interface

Encourage Saudi citizens to be active and connect through sport, without friction.

Protected user accounts

Keep user information safe — its integrity and its confidentiality.

CONSTRAINTS

Language

JavaScript, building on the prototype stack: HTML, CSS, Vite, Supabase, Vercel.

Time

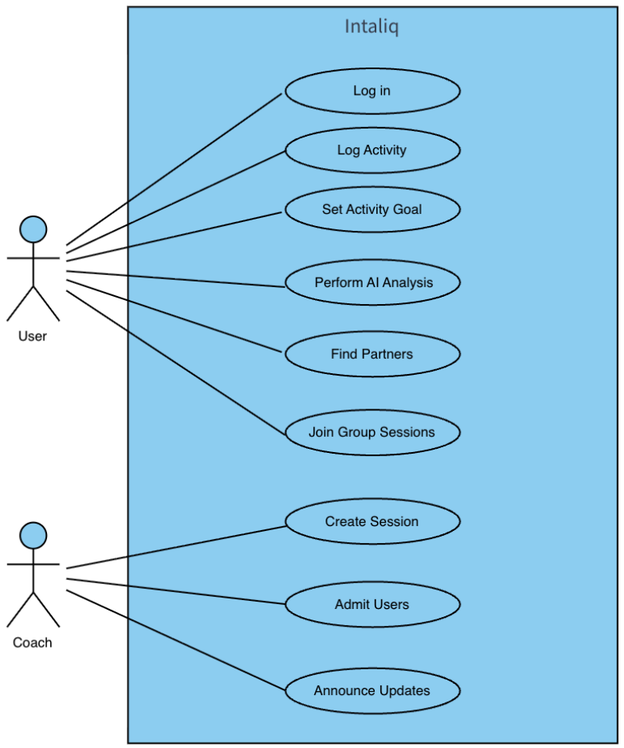
Four fixed milestones between 30 Sep and 4 Nov 2026.

Cost

Limits the services and hosting the app can rely on.

SAD 4.1 · USER VIEW · USE-CASE REALIZATION

Two actors, nine use cases



User

6 USE CASES

- Log in
- Set activity goal
- Find partners
- Log activity
- Perform AI analysis
- Join group sessions

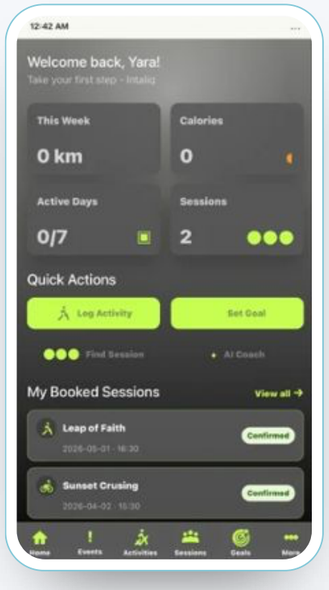
Coach

3 USE CASES

- Create session
- Announce updates
- Admit users

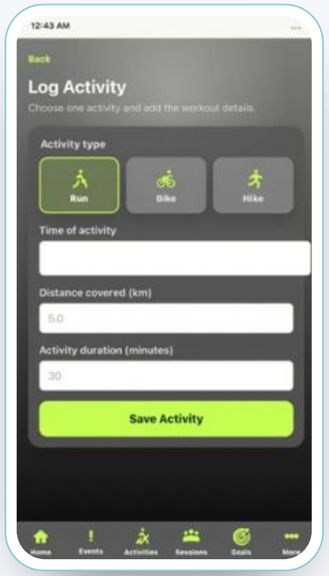
USER | COACH

The use cases, on screen



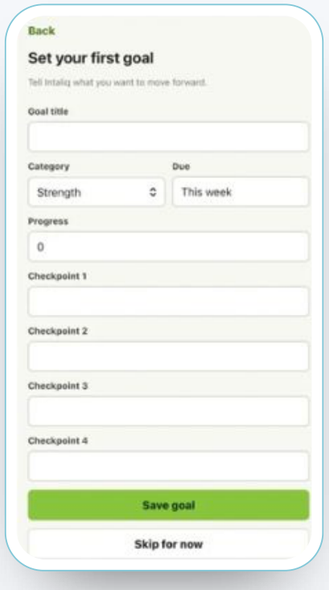
HOME

Dashboard



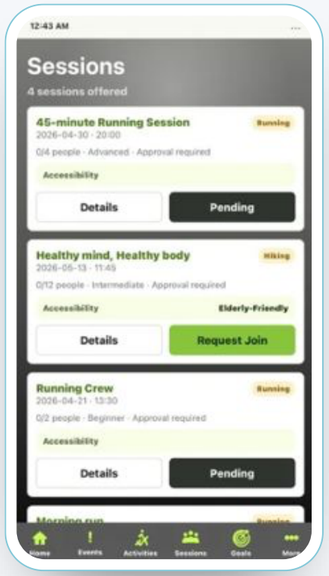
USE CASE

Log activity



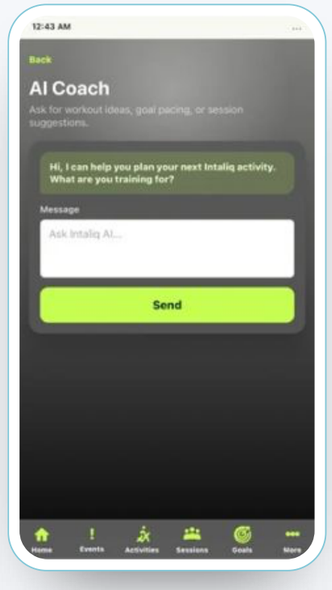
USE CASE

Set activity goal



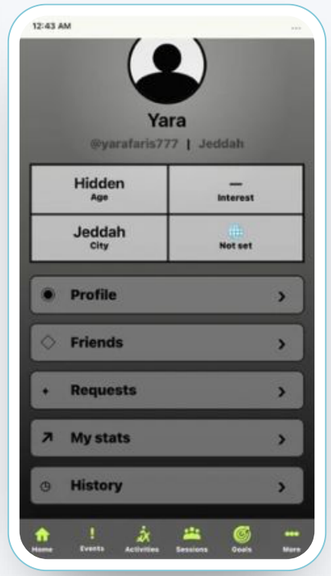
USE CASE

Join group sessions



USE CASE

AI analysis

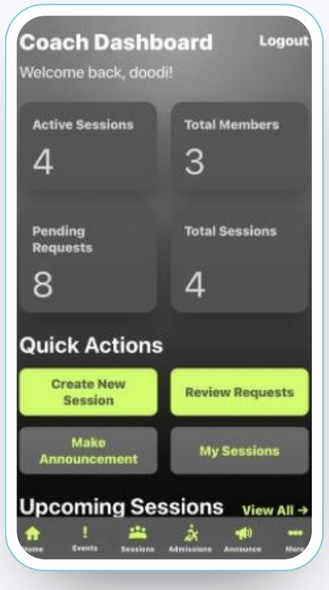


USE CASE

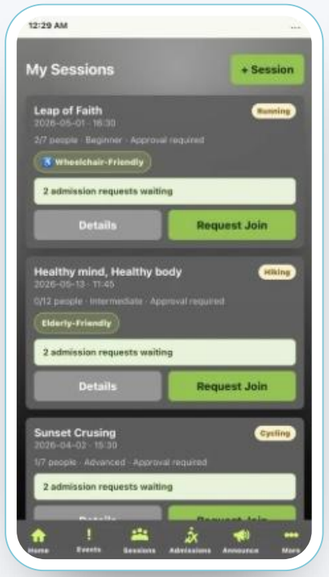
Find partners

USER COACH

The use cases, on screen



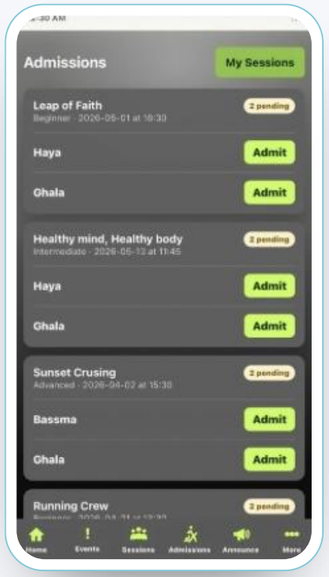
HOME
Coach dashboard



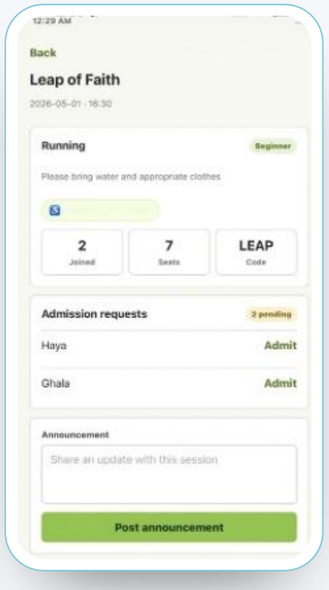
OVERVIEW
My sessions



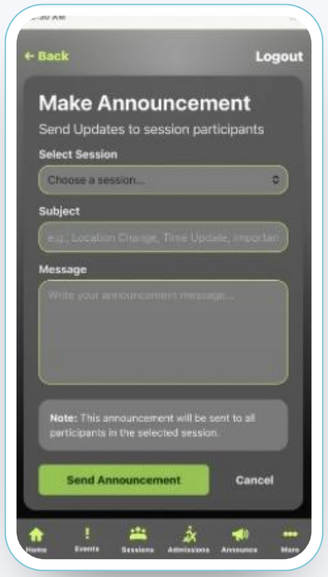
USE CASE
Create session



USE CASE
Admit users



USE CASE
Session detail



USE CASE
Announce updates

SAD 4.2 · SCENARIOS

Three scenarios, end to end

BOOK A SESSION | AI CHATBOX | ACTIVITY GOAL

ACTOR

User

STIMULUS

User selects and books a session

RESPONSE

Request submitted; user added if the coach accepts, notified if not

NORMAL SCENARIO

1. Open the Intaliq app
2. Log in
3. Browse available group sessions
4. Select a session
5. Review session and coach details
6. Submit a booking request
7. Coach reviews the request
8. Coach accepts the request
9. System confirms and notifies the user

ALTERNATIVE SCENARIOS

- **Not logged in** → login pop-up
- **Session full** → request blocked with a pop-up
- **Coach rejects** → user is notified
- **System error** → request not completed, user notified

Sequence diagram

SAD 4.2 · SCENARIOS

Three scenarios, end to end

BOOK A SESSION | AI CHATBOX | ACTIVITY GOAL

ACTOR

User

STIMULUS

User sends a query in the AI chat

RESPONSE

The AI answers and suggests further help

NORMAL SCENARIO

1. Open the Intaliq app
2. Log in
3. Select the “More” tab
4. Select AI Chatbox
5. Pick a suggested prompt or write one
6. Submit the message
7. AI coach answers the query
8. User accepts the answer and closes the chat

ALTERNATIVE SCENARIOS

Not logged in → login pop-up

Medical advice requested → chatbox refuses and ends the chat; only specialists should advise

System error → chat request not completed, user notified

SAD 4.2 · SCENARIOS

Three scenarios, end to end

BOOK A SESSION | AI CHATBOX | ACTIVITY GOAL

ACTOR
User

STIMULUS
User enters an activity goal

RESPONSE
The system confirms the goal has been saved

NORMAL SCENARIO

1. Open the Intaliq app
2. Log in
3. Select “Activity”
4. Select “Activity Goal”
5. Enter the goal — activity type and target
6. Submit the goal
7. System validates and saves the goal
8. System confirms it was created

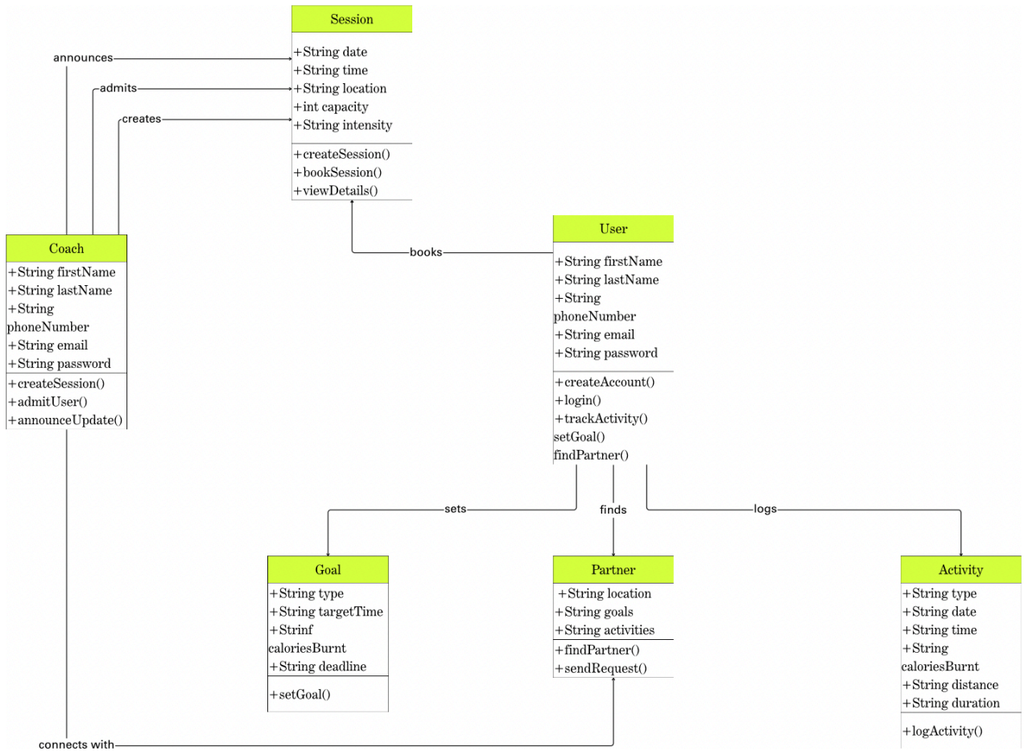
ALTERNATIVE SCENARIOS

- **Not logged in** → login pop-up
- **Missing field** → prompted to complete it
- **Invalid data** → error message shown
- **System error** → goal is not saved

Sequence diagram | Activity diagram

SAD 5.1 · LOGICAL VIEW · CLASS DIAGRAM

Six classes model the domain



COACH → SESSION

creates, admits into, announces

USER → SESSION

books a group session

USER → GOAL · ACTIVITY

sets goals, logs activities

USER · COACH → PARTNER

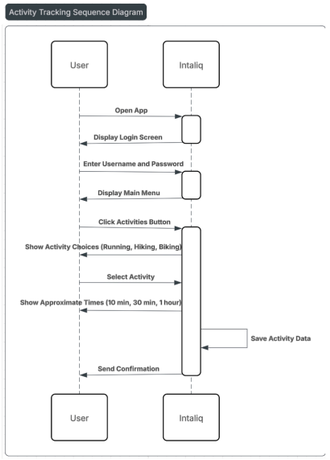
find and connect with partners

SAD 5.2 · LOGICAL VIEW · SEQUENCE DIAGRAMS

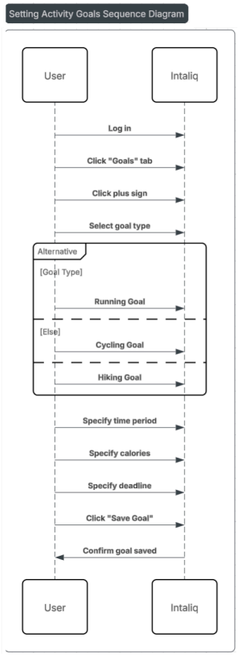
Five interactions, message by message

Click any diagram to open it full screen.

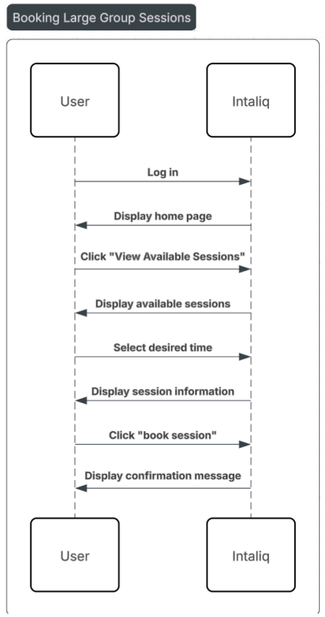
LOG ACTIVITY



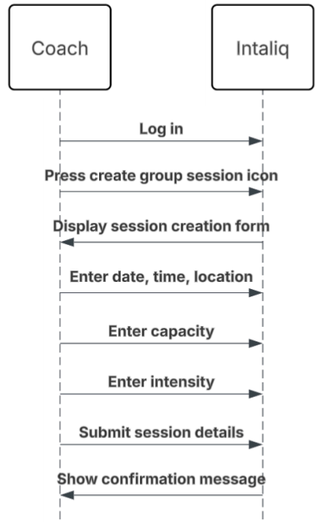
SET GOAL



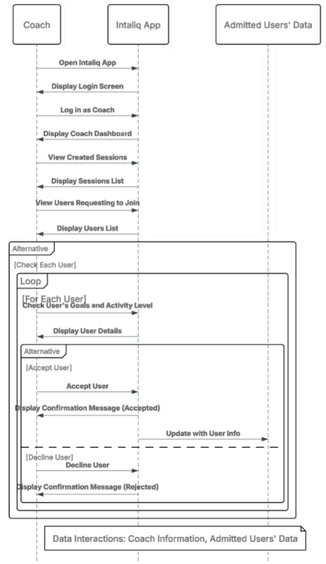
BOOK SESSION



CREATE SESSION



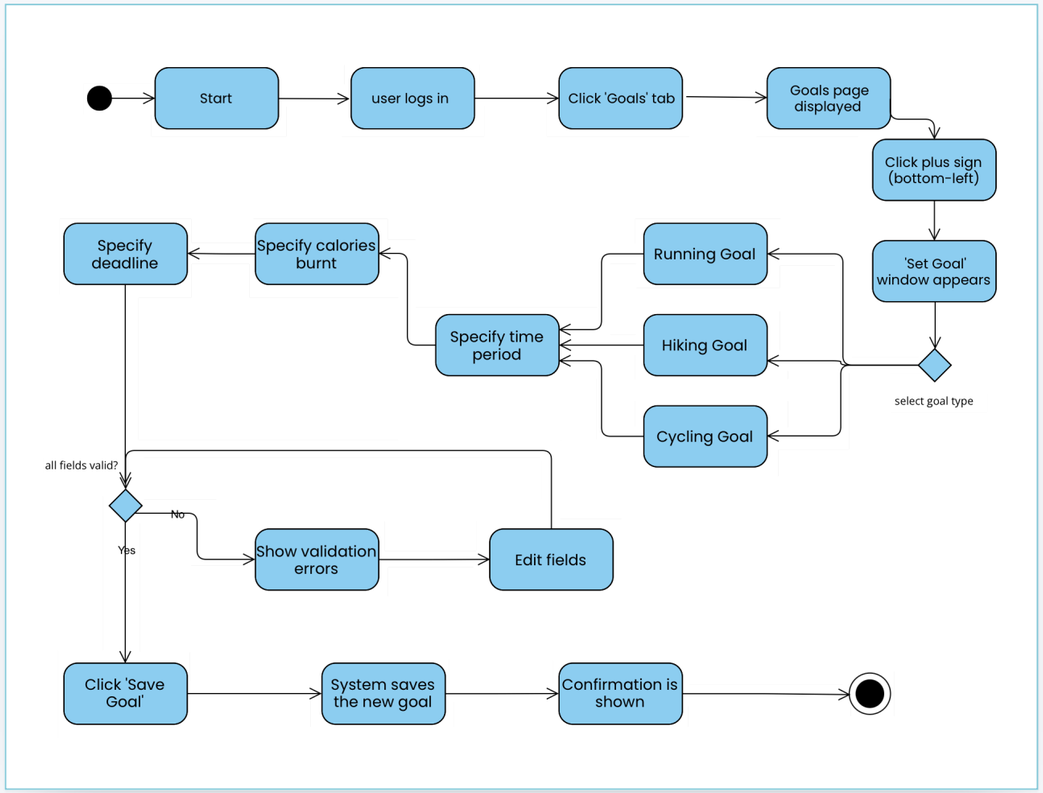
ADMIT MEMBERS



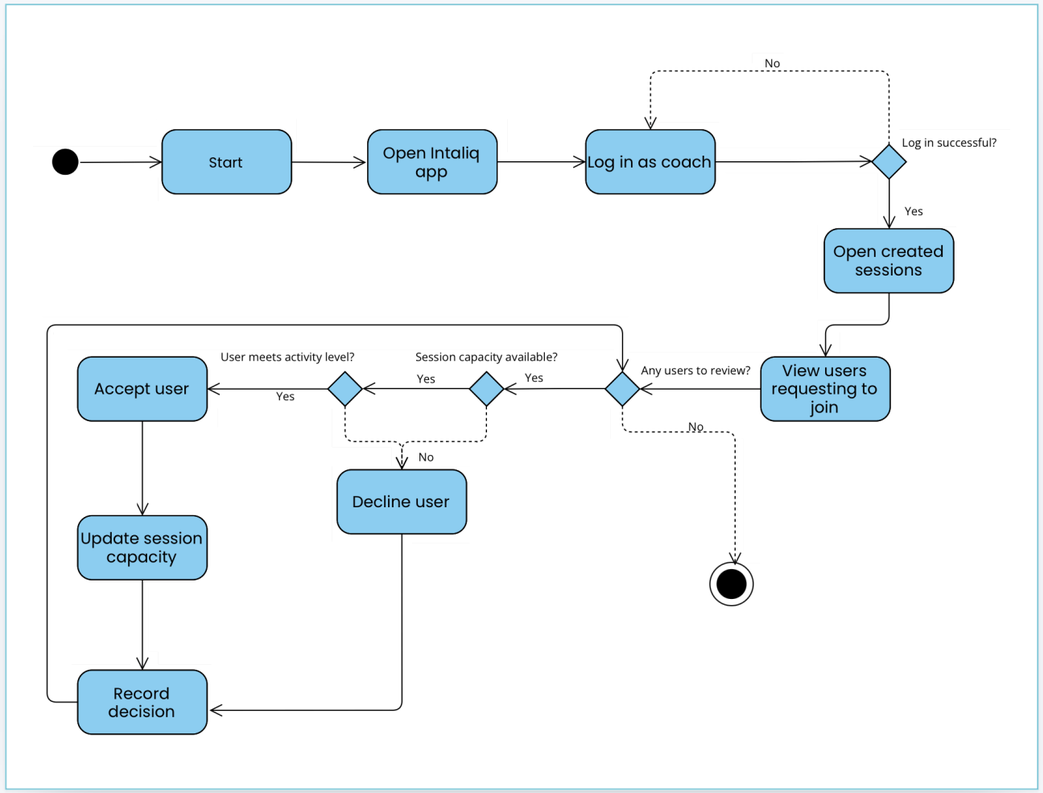
SAD 6.1 · PROCESS VIEW · ACTIVITY DIAGRAMS

How the work flows at runtime

SETTING AN ACTIVITY GOAL · USER

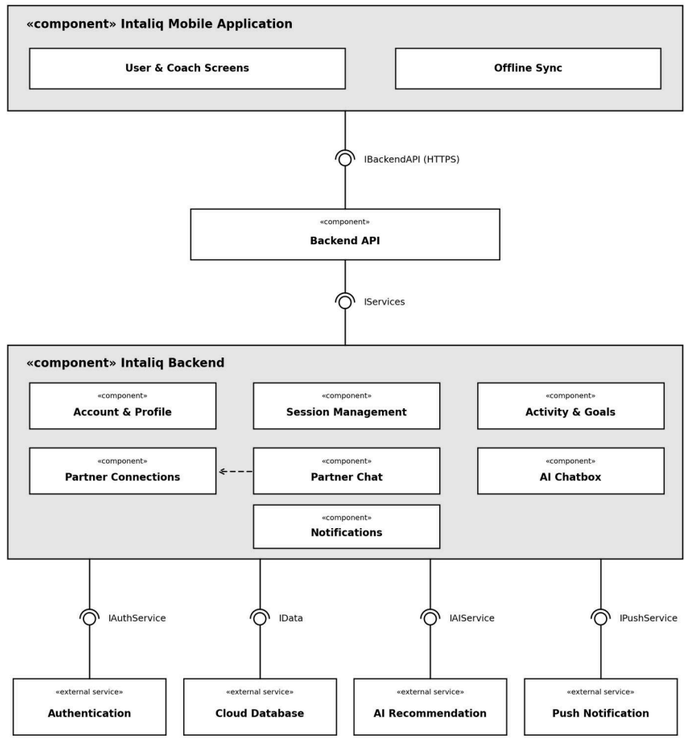


ADMITTING MEMBERS · COACH



SAD 7.1 · DEVELOPMENT VIEW · COMPONENT DIAGRAM

Components and the interfaces between them



Mobile application

User & coach screens, plus Offline Sync, which holds actions until the device is back online.

Backend API · `IBackendAPI (HTTPS)`

Account & profile, sessions, activities & goals, partner connections, partner chat, AI chatbox and notifications.

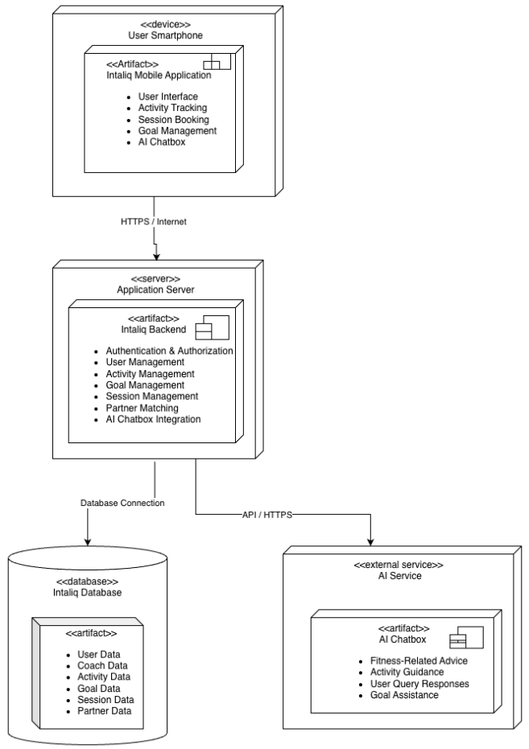
External services

Authentication, cloud database, AI recommendation and push notifications — each behind its own interface.

ⓘ Partner Chat is only reachable once a Partner Connection exists.

SAD 8.1 · PHYSICAL VIEW · DEPLOYMENT DIAGRAM

Where the software runs



«DEVICE»

User smartphone

Runs the Intaliq mobile app: UI, tracking, booking, goals, AI chat.

«SERVER»

Application server

Intaliq backend: auth, users, activities, goals, sessions, matching.

«DATABASE»

Intaliq database

User, coach, activity, goal, session and partner data.

«EXTERNAL SERVICE»

AI service

Fitness advice, activity guidance and goal assistance.

PHONE → SERVER · HTTPS

SERVER → DB · CONNECTION

SERVER → AI · API / HTTPS

DELIVERABLE 2 · 20%

Architecture Documentation

✓ SUBMITTED 10 OCT

QUALITY ATTRIBUTES · STYLE

4	4	1	4
quality attributes	measurable scenarios	architectural style	style components

02

D2 1 · QUALITY ATTRIBUTES

The four qualities Intaliq is built around

Security

Accounts, logs and personal goals are private. Every user and coach is authenticated and limited to their own functions.

100%

UNAUTHORISED ATTEMPTS REJECTED

Availability

Users rely on the app to track, book and connect. Failures are reported, and Offline Sync holds actions until reconnection.

100%

QUEUED ACTIVITIES SYNCED

Performance

Checking sessions, logging activity and recording goals should happen without noticeable delay.

95% < 3s

SESSION-LIST REQUESTS

Usability

The goal is motivation: every feature is reachable without complex steps, with alerts confirming each action.

90%

FIRST-TIME USERS SUCCEED UNAIDED

D2 2 · QUALITY ATTRIBUTE SCENARIOS

SECURITY | AVAILABILITY | PERFORMANCE | USABILITY

Each quality, made testable

SOURCE

An unauthorised user

STIMULUS

Tries to open another user's fitness data without authenticating

ENVIRONMENT

ARTIFACT

Authentication & authorization component

Normal system operation

RESPONSE

Verifies identity and permissions, then denies access

RESPONSE MEASURE

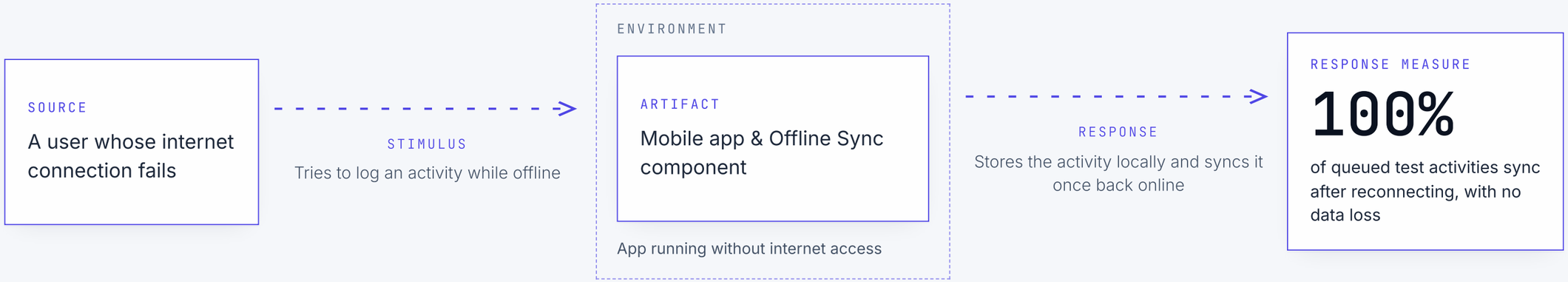
100%

of unauthorised attempts in the security test cases rejected

D2 2 · QUALITY ATTRIBUTE SCENARIOS

SECURITY | AVAILABILITY | PERFORMANCE | USABILITY

Each quality, made testable



D2 2 · QUALITY ATTRIBUTE SCENARIOS

SECURITY | AVAILABILITY | **PERFORMANCE** | USABILITY

Each quality, made testable

SOURCE
An authenticated user

STIMULUS
Asks to view available group sessions

ENVIRONMENT

ARTIFACT
Mobile app, backend API & database

Normal operation, active connection

RESPONSE
Retrieves the sessions and displays their details

RESPONSE MEASURE

95% • 3s

of session-list requests complete within 3 s under normal load

D2 2 · QUALITY ATTRIBUTE SCENARIOS

SECURITY | AVAILABILITY | PERFORMANCE | USABILITY

Each quality, made testable

SOURCE

A new Intaliq user

STIMULUS

Tries to create a personal fitness goal

ENVIRONMENT

ARTIFACT

Mobile app user interface

Normal app operation

RESPONSE

Guides activity choice, goal entry and saving, then confirms

RESPONSE MEASURE

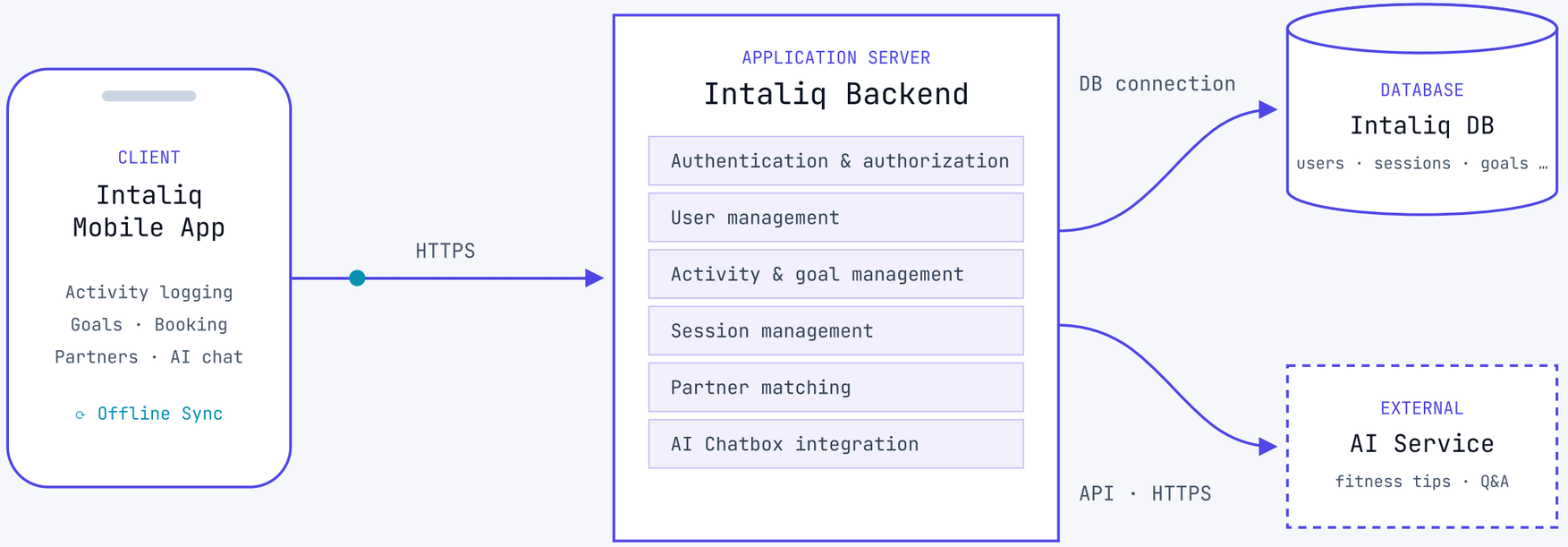
≥ 90%

of first-time test users create a goal without assistance

D2 3.1 & 3.3 · ARCHITECTURAL STYLE

Client–server, with a central backend

The mobile app is the client. One backend handles authentication, business logic and every call to the database and the AI service.



D2 3.2 & 3.3 · JUSTIFICATION & COMMUNICATION

Why this style fits Intaliq

Data centralization
Users, actions, goals and sessions controlled through one backend and database.

Security
Authentication and authorization enforced on the server.

Maintainability
App and backend evolve independently.

Performance
Heavy processing stays on the server; the app focuses on the user.

Integration
Clean exchange with third-party services such as the AI chatbox.

COMMUNICATION PROCESS

1. User performs an action in the mobile app
2. App sends the request to the backend over HTTPS
3. Backend authenticates and processes it
4. Backend calls the database or AI service if needed
5. Backend returns the outcome to the app
6. App shows the result to the user

DELIVERABLE 3 · 20%

Software Design Documentation

✓ ISSUED 25 OCT

GANG OF FOUR PATTERNS

5	3	3	5
design concerns	design views	design patterns	new diagrams

03

SDD 2.2 – 2.3 · STAKEHOLDERS & DESIGN CONCERNS

Every decision answers someone's concern

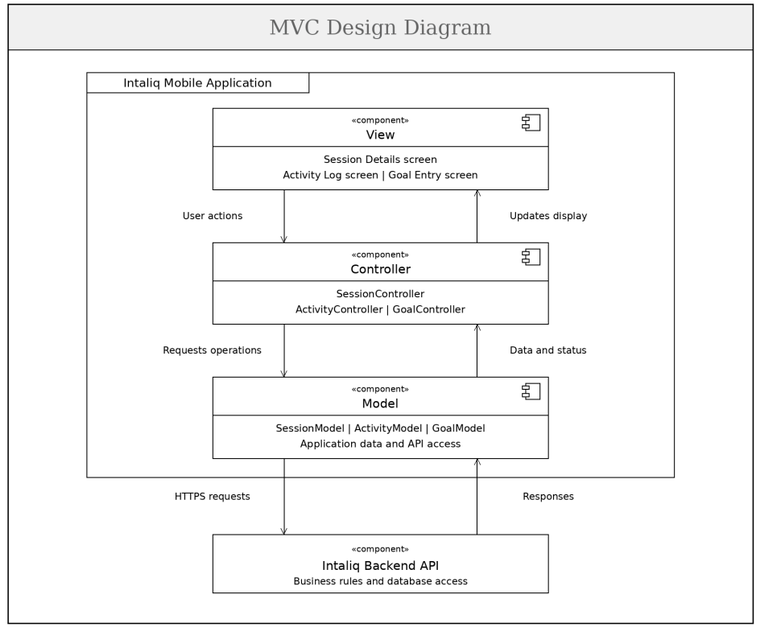
End users, coaches, developers, testers and maintainers each need something different from the design.

01	DEVELOPERS · MAINTAINERS Add new activity types without changing existing code	→ FACTORY METHOD
02	END USERS · COACHES Get timely updates about sessions and booking decisions	→ OBSERVER
03	DEVELOPERS · PRODUCT OWNERS Use external services without depending on one provider	→ ADAPTER
04	TESTERS Test each part of the system on its own	→ CLIENT-SERVER · LAYERS
05	END USERS Keep personal data secure	→ DATA ONLY BEHIND THE SERVER

SDD 2.4 – 2.6 · ARCHITECTURAL VIEWPOINTS

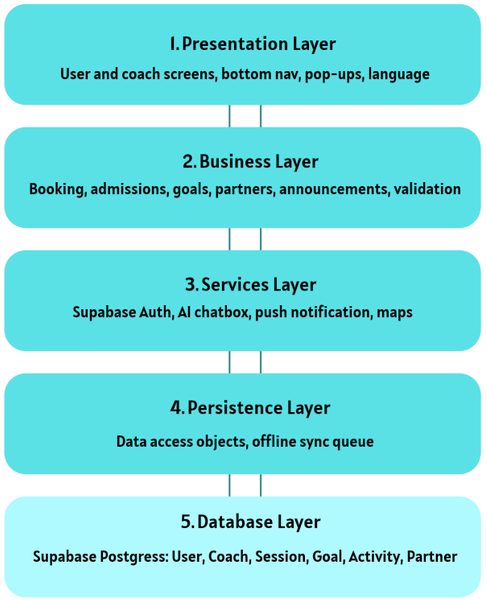
Inside the client, and inside the server

MOBILE APP · MVC



View shows screens, **Controller** handles actions, **Model** holds data and calls the backend API — easier to maintain, extend and test.

BACKEND · LAYERED



Each layer talks only to its neighbour, so changes stay local, the database is reached only through the layers above it, and each layer is tested alone.

SDD 3.1 · DETAILED DESIGN

Three Gang of Four patterns, one of each kind

DP-01 · CREATIONAL

Factory Method

Creates the right activity — running, hiking or cycling — without the logging process knowing which.

ACTIVITY LOGGING | CONCERN 01

DP-02 · BEHAVIOURAL

Observer

Notifies users and coaches automatically when a session changes or a booking is decided.

NOTIFICATIONS | CONCERN 02

DP-03 · STRUCTURAL

Adapter

Lets the AI chatbox talk to an external AI service through one standard interface.

AI CHATBOX | CONCERN 03

DP-01 · FACTORY METHOD · CREATIONAL

One logging process, any activity

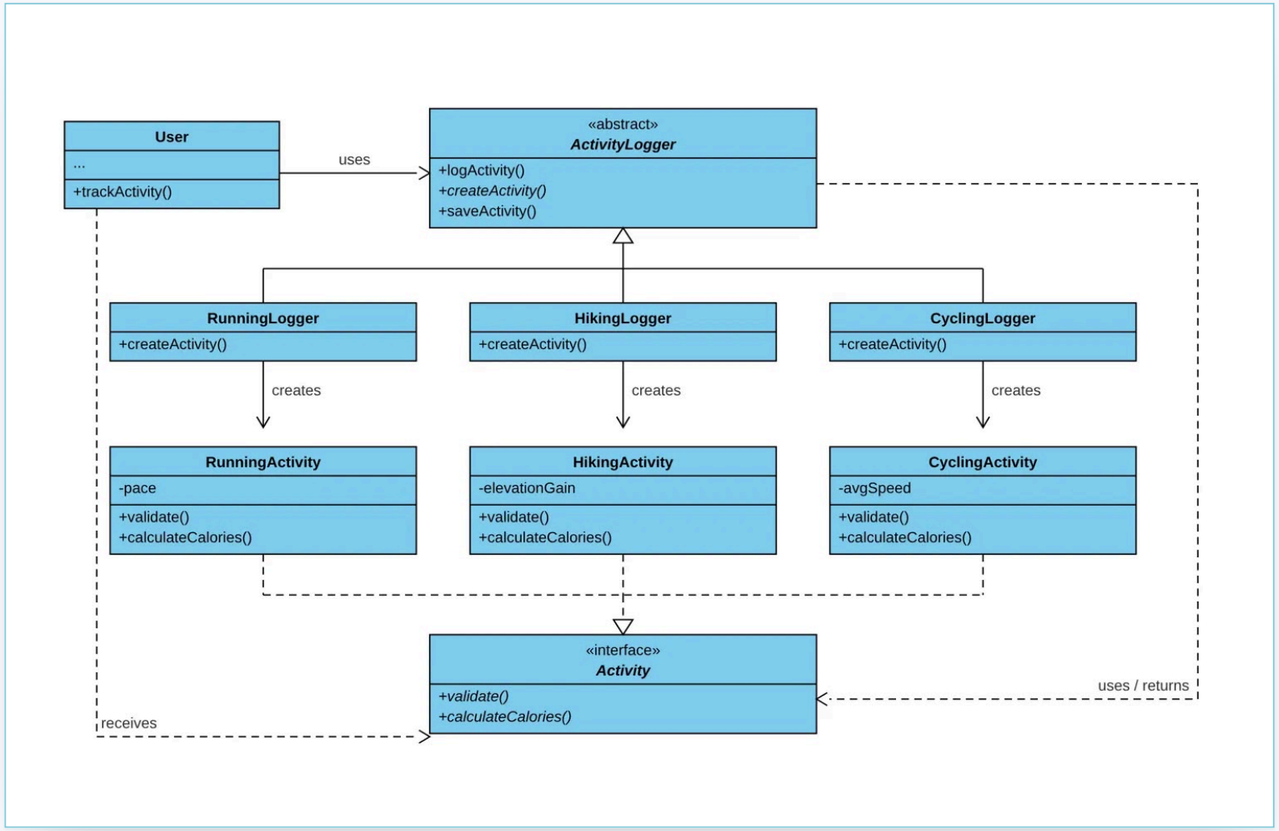
HOW IT'S APPLIED

RunningLogger, HikingLogger and CyclingLogger each override `createActivity()`. Pick hiking, and the HikingLogger builds a HikingActivity that the logging process uses without knowing its type.

WHY FACTORY METHOD

Every activity is validated and saved the same way but counts calories differently. A new sport such as padel only needs its own logger and activity class — nothing else changes.

Log activity · sequence



DP-02 · OBSERVER · BEHAVIOURAL

One change, everyone informed

HOW IT'S APPLIED

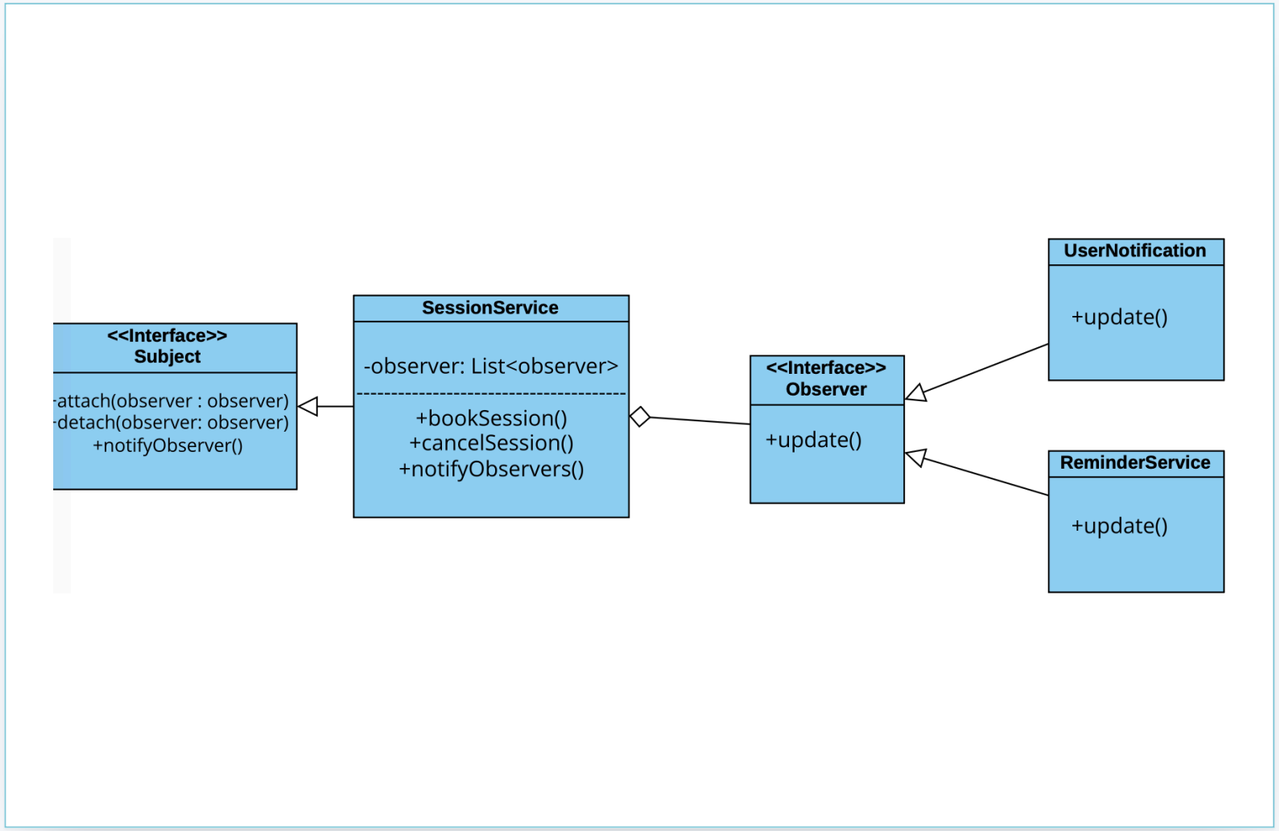
SessionService is the subject; UserNotification and ReminderService observe it. When a class is cancelled, each observer reacts on its own — one alerts the user, the other reschedules reminders.

WHY OBSERVER

Announcements reach every admitted member, and booking decisions reach the requester (Book a Session). Session never handles delivery, so new channels such as email plug in without changing it.

Book session · sequence

Admit members · sequence



DP-03 · ADAPTER · STRUCTURAL

Any AI provider, one interface

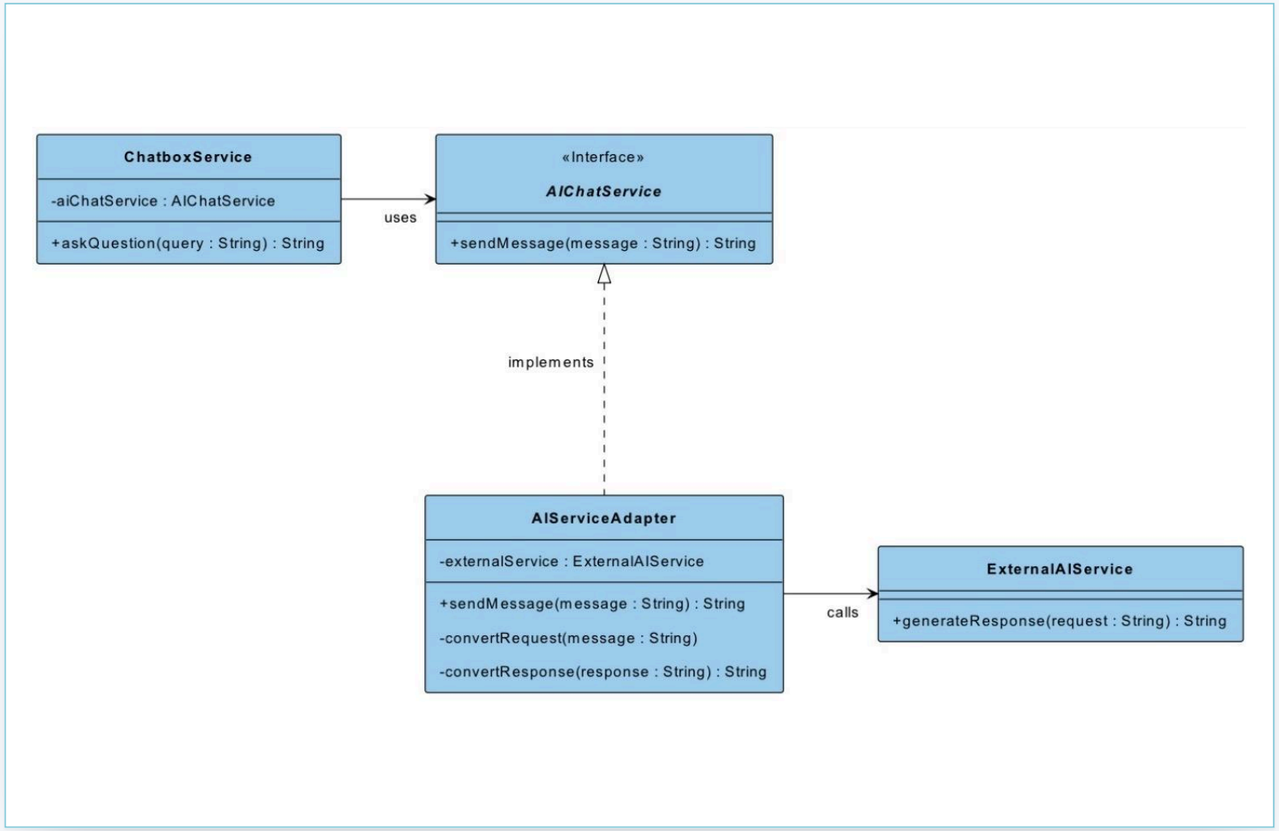
HOW IT'S APPLIED

ChatboxService only calls `AIChatService.sendMessage()`. AIServiceAdapter converts the question into the external service's format, sends it over HTTPS, and converts the answer back.

WHY ADAPTER

Provider-specific formats stay out of the chatbox logic. If the external API changes, only the adapter changes; switching provider means writing a new adapter.

Component diagram



DELIVERABLE 4 · 20%

Validation & Presentation

DUE 4 NOV

REVIEW EVERY DESIGN DECISION

SRS
traced to every view

SE365
usability findings applied

17
course requirements checked

04

VALIDATION · REQUIREMENTS COVERAGE

Does the design meet every requirement?

D1 · SAD OUTLINE — 4+1

1 · Introduction (overview → references)	COVERED →
2 · Architectural representation	COVERED →
3 · Goals & constraints	COVERED →
4 · User view — use cases + ≥ 3 scenarios	3 OF 3 →
5a · Class diagram	COVERED →
5b · Collaboration diagram	OPTIONAL –
5c · Sequence diagrams	5 DRAWN →
6 · Process — activity diagrams	2 DRAWN →
7 · Development — component diagram	COVERED →
8 · Physical — deployment diagram	COVERED →

D2 · ARCHITECTURE — QA + STYLE

3–4 quality attributes	4 →
Scenarios drawn & documented	4 →
Style chosen, justified, implemented	COVERED →

D3 · DESIGN — SDD

≥ 3 design patterns	3 →
Justification for each	COVERED →
Software Design Document (SDD)	ISSUED →

D4 · VALIDATION — REVIEW

Design document reviewed & validated	PENDING →

VALIDATION · REVIEW LOG

Checked against the SRS and the prototype

REVIEW CHECKLIST

- Every 4+1 view traced to SRS features FR-01 – FR-07 — DONE
- SE365 prototype usability findings folded into design decisions — DONE
- Trade-off: security checks vs. unaided first-time goal creation — IN PROGRESS
- Peer review of views and patterns with the instructor — PENDING

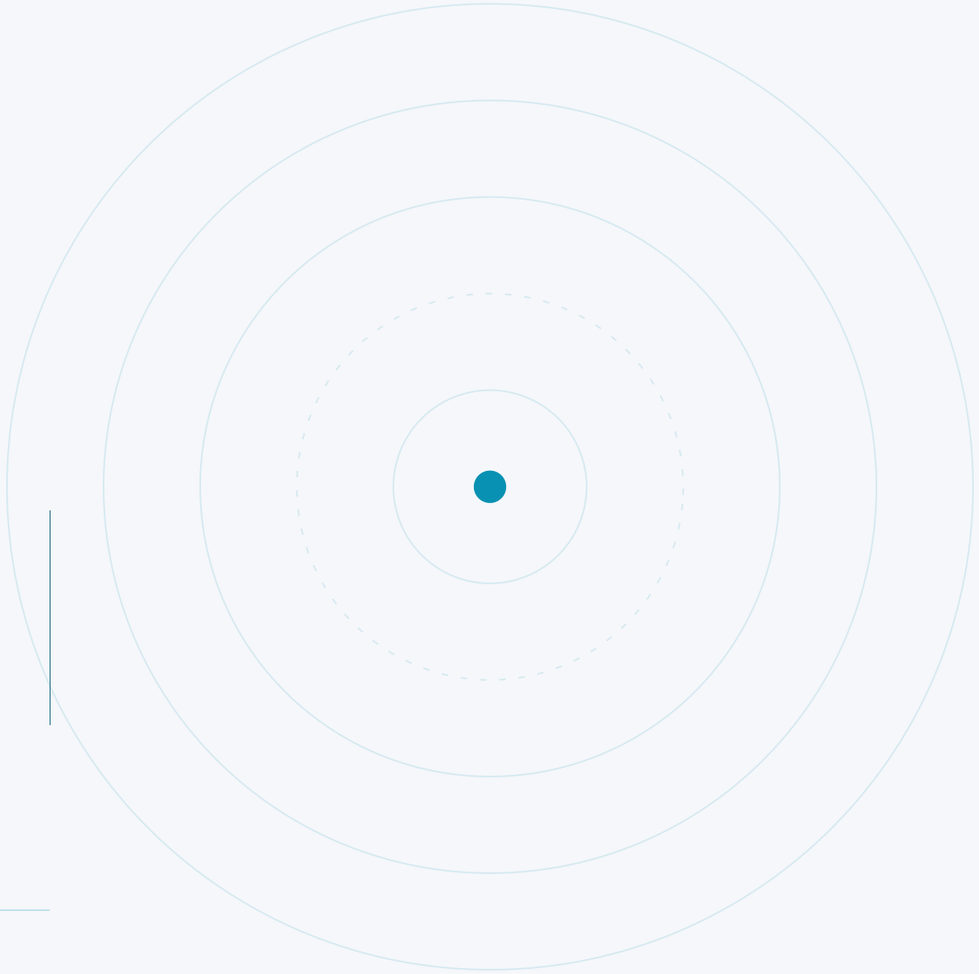
NEXT STEPS

1. Check the SDD's patterns against the SRS and Deliverables 1–2
2. Close the security ↔ usability trade-off
3. Instructor review of views and patterns
4. Final presentation and validation — 4 Nov

INTALIQ · SE322

Thank you

Questions?



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